Overview

Understand the architecture and capabilities of VCF across the stages in the private cloud lifecycle.

In This Chapter

- What Is VMware Cloud Foundation?
- What Is vSphere Foundation?
- VCF Taxonomy
- VCF Capabilities by Job Function
- Getting Started with VMware Cloud Foundation
- Getting Started with vSphere Foundation

Other Resources for Getting to Know VCF

In addition to an overview of the VCF capabilities, see VCF Release Notes and Overview of VCF Deploy, Converge, and Upgrade.

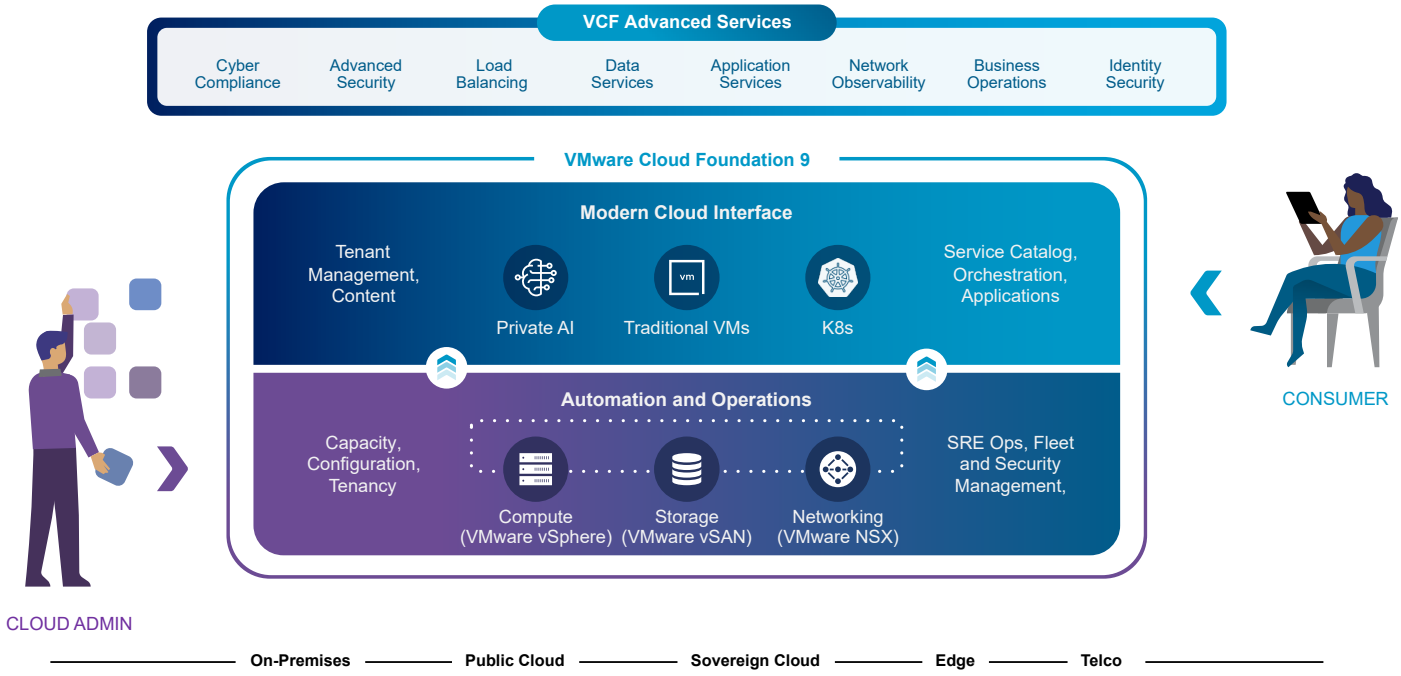
What Is VMware Cloud Foundation?

VMware Cloud Foundation ® (VCF) is a private cloud platform that delivers public cloud scale and agility with on-premises security, resilience and performance, while lowering total cost of ownership.

VMware Cloud Foundation can help you accelerate digital innovation with faster infrastructure modernization, a unified cloud experience, along with stronger cyber resilience and platform security.

VCF delivers a consistent private cloud experience across any environment whether it is a customer-managed data center, edge location, or any cloud endpoint. VCF supports both traditional as well as modern workloads such as containers and AI/ML workloads, and brings modern applications faster to the market with self-service Infrastructure as a Service (IaaS) consumption experience for the application teams.

Figure 1: Modern Private Cloud Platform with VMware Cloud Foundation



- VMware Cloud Foundation is a full-stack IaaS platform that delivers software-defined compute, storage, networking, security, and management.
- Integrated automation enables a self-service platform to deploy VMs and containers for developer agility.
- Hardened platform offering built-in resilience, scaling, and clustering for non-stop operations.
- Provides cloud agility to scale infrastructure without scaling staff, delivering cloud consumption on premises.
- Provides automation and orchestration to simplify Day 0, Day 1, and Day 2 tasks.
- Simplifies full stack deployment.

What's New Information

For information about what's new in VCF 9.1, see Release Notes.

VCF Components

VCF consists of the following components:

- VCF Installer
- vSphere
- vSAN
- NSX
- VCF Operations
- VCF Automation

Getting Started and a List of Capabilities

To learn how to get started, see Getting Started with VMware Cloud Foundation.

For a detailed list of capabilities and links to the documentation for them, see VCF Capabilities by Job Function.

VCF Installer Overview

VCF Installer is a dedicated virtual machine that helps you plan, configure, and deploy VCF and vSphere Foundation components.

Use the VCF Installer to perform the following operations:

- Download the binaries to deploy VCF or vSphere Foundation.
- Plan the desired size and configuration of your VCF or VMware vSphere Foundation deployment.
- Deploy and configure a new VCF or VMware vSphere Foundation platform. You can also reuse existing VCF Operations and VCF Automation instances for new VCF fleets.
- Use existing virtual infrastructure to converge them to a VCF or vSphere Foundation platform.
- Extend an existing VCF fleet by deploying an additional VCF Instance.
- Deploy VCF management services and license server to complete the vSphere Foundation upgrade to 9.1.
- Review historical deployment process details, download passwords, and download JSON specifications from VCF or vSphere Foundation deployments.

VCF Installer works with major, minor, and maintenance releases, but does not support patch or express patch releases. After the VCF Installer workflows complete, you can apply patches manually.

VCF Installer is part of the SDDC Manager appliance and the binaries for both reside in the `VCF-SDDC-Manager-Appliance-9.x.x.ova` file.

VCF Installer Modes

For VCF deployments, depending on where you deploy VCF Installer, it can deploy multiple VCF platforms, or it can become the SDDC Manager instance for the target VCF Instance that you deploy.

VCF Installer mode	If you deploy the VCF Installer appliance on an ESX host that is not part of the management domain for the target deployment, then the appliance stays in installer mode after the deployment completes. In this mode, you can deploy additional VCF Instances and perform convergence workflows.
VCF Installer to SDDC Manager mode	If you deploy the VCF Installer appliance on a host that is part of the management domain, during the deployment, the appliance switches to SDDC Manager mode. Once in SDDC Manager mode, the appliance can no longer work in installer mode. As a best practice, deploy the appliance on a local datastore, then during the deployment workflows, the appliance migrates to the primary datastore. For FC storage, the appliance can initially be deployed on the FC datastore.

VCF Installer Documentation

For more information about the VCF Installer and the options for deployment, convergence, and upgrade to VCF and vSphere Foundation it provides, see Deployment, Convergence, and Upgrade.

vSphere Overview

Virtualize physical infrastructure into managed CPU, storage, and networking resources.

The two core components of VMware vSphere® are VMware ESX® and VMware vCenter®. ESX is the virtualization platform where you create and run virtual machines and virtual appliances, Kubernetes workloads, and AI/ML workloads. vCenter is the service through which you manage multiple hosts connected in a network and pool host resources.

vSphere manages large collections of infrastructure, such as CPUs, storage, networking, GPUs, and DPUs as a seamless and dynamic operating environment, and also manages the complexity of a data center.

Lifecycle Management of vSphere

- Lifecycle management is achieved using pre-staging ESX images, remediating hosts in parallel, and by applying updates in parallel across clusters.
- vSphere can manage infrastructure images to patch, update, or upgrade clusters using a desired state model.
- vSphere can reduce maintenance windows using reduced downtime upgrades for vCenter and live patching for ESX for near zero downtime for security patches.

Workload Performance

- With its GPU support, vSphere improves workload performance for modern AI workloads
- vSphere on DPUs (VMware vSphere® Distributed Services EngineTM) helps accelerate infrastructure network functions on the Data Processing Unit (DPU) by using available CPU cycles to achieve higher workload consolidation per host.
- Distributed Resource Scheduler (vSphere DRS) enables automatic load balancing of resources that are allocated to workloads in a vSphere cluster. Storage DRS optimizes VM data placement as the VM is created and used over time.

Business Continuity

- High Availability with vSphere automatically restarts your VMs following physical machine failure.
- Fault Tolerance provides continuous availability of any application if a hardware failure with no data loss or downtime occurs.
- vMotion enables live migration of virtual machines with no disruption to users or loss of service, eliminating the need to schedule application downtime for planned server maintenance. Storage vMotion avoids downtime for planned storage maintenance.

Built-in Security

- Identity federation with Microsoft Entra ID (formerly Azure AD), ADFS, and Okta: Secure access and account management.
- Virtual Machine Encryption: Data-at-rest encryption for virtual machine data and disks.
- vSphere Trust Authority: Remote attestation for sensitive workloads.

vSphere Documentation

For information about the built-in vSphere capabilities, see the vSphere documentation.

vSAN Overview

VMware® vSAN™ aggregates local or direct-attached capacity devices of all ESX hosts in a cluster and creates a single storage pool that is shared across all ESX hosts in the vSAN cluster.

You can use vSphere features that require shared storage, such as vSphere HA, vSphere vMotion, and vSphere DRS, without the need for external shared storage.

Performance Across Workloads

vSAN delivers high throughput and low latency storage. For best performance, you can use vSAN Express Storage Architecture™ (vSAN ESA) with high performance NVMe-based TLC flash devices.

Agile Scaling for Deployments of All Sizes

vSAN offers flexible scaling ranging from clusters as small as two nodes (plus witness) to clusters with up to 64 nodes. Whether scaling out by adding nodes to a cluster, scaling up by adding drives to nodes or scaling compute and storage resources independently with disaggregation, vSAN enables you to customize performance and capacity to optimize storage resources and drive down costs. Disaggregation in vSAN decouples compute and storage resources. As a result, compute and storage resources can be scaled independently within your environment. Disaggregation enables the use of dedicated storage-only clusters.

Unified Operations and Lifecycle Management

vSAN provides operational consistency from edge to cloud, with complete visibility and lifecycle management of compute and storage from a single UI.

Storage Capacity and Efficiency

Improve storage efficiency with optimized compression methods. A single-tier architecture option eliminates the need for a cache tier, further enhancing storage efficiency. vSAN enables you to store data using RAID 5 or RAID 6 for efficiency with performance at the speed of RAID 1.

Integrated File Services

vSAN enables provisioning of file shares with a single workflow and provides a unified storage control plane for both block and file storage. vSAN file services can integrate Active Directory and support Kerberos network authentication and the most common protocols. vSAN file services can be used in two-node deployments and stretched cluster deployments.

VM-Centric Policy-Based Management

You can perform consistent, VM-centric operations by using policy-based management. Within the vSAN ESA, per-VM policies increase flexibility of management, enabling compression policies, or customize data services on a per-VM basis.

vSAN Documentation

For more information about the built-in vSAN capabilities, see the vSAN documentation.

NSX Overview

VMware NSX® simplifies network connectivity, consumption, and operations with scalable multi-site connectivity, efficient infrastructure utilization, and self-service for application teams. It uses a software-defined approach to networking that spans data centers, clouds, and application frameworks.

NSX brings networking services closer to the application, whether running on virtual machines (VMs), containers, or physical servers. This allows for a cloud operating model where networks can be provisioned and managed independently of the underlying hardware.

NSX replicates the entire network model in software, enabling any network topology—from simple to complex multi-tier networks—to be created and provisioned in seconds. Users can create multiple virtual networks with diverse requirements, leveraging a combination of the services offered over NSX or from a broad ecosystem of third-party integrations to build inherently more agile and secure environments. These services can then be extended to various endpoints within and across clouds.

Layer 2 - Layer 7 Network Stack and Overlay Services

NSX provides a complete set of logical networking capabilities and services, including logical switching, routing, distributed load balancing, virtual private network (VPN), quality of service (QoS), and monitoring. These services are

provisioned in virtual networks. Virtual networks are deployed non-disruptively over the networking hardware and can extend across data centers, private clouds, and container platforms.

- Switching. Enable logical Layer 2 overlay extensions across a routed (Layer 3) fabric within and across data center boundaries.
- Routing. Dynamic routing between virtual networks that is performed in a distributed manner in the hypervisor kernel, and scale-out routing with active-active failover with physical routers. Static routing and dynamic routing protocols are supported, including support for IPv6.
- Virtual routing and forwarding (VRF). Complete data plane isolation among tenants with a separate routing table, network address translation (NAT), and edge firewall support in each VRF on the NSX Tier-0 gateway.
- NSX gateway. Support for bridging between VLANs configured on the physical network and NSX overlay networks, for seamless connectivity between virtual and physical workloads.
- VPN. Site-to-site and unmanaged VPN for cloud gateway services.

Multi-Site Networking and Disaster Recovery

By using NSX, you can move workloads between racks and data centers by decoupling them from the underlying networks. You can use network overlays and a consistent policy framework across different infrastructures without the need for any changes to the network when moving workloads. This capability supports workload mobility for disaster recovery and workload balancing, ensuring continuous availability and optimal application performance.

- Federation. Centralized policy configuration and enforcement across multiple locations from a single pane of glass, enabling network-wide consistent policy, operational simplicity, and simplified disaster recovery architecture.
- Multi-cloud networking. Enable consistent networking and security across data center sites, and across private cloud boundaries, irrespective of underlying physical topology or cloud platform.

Network Automation with Declarative APIs

NSX can manage the provisioning, deployment, operations, and retirement of software defined networking infrastructure and applications from a central control pane. You can automate infrastructure operations and offload manual tasks with workflows and agile templating using tools such as VCF Automation, Terraform, and Ansible. NSX abstracts network services from the underlying hardware, enabling programmable infrastructure through APIs. You can implement policies and workflows to automate tasks like network provisioning, security enforcement, load balancing, streamlining operations, and improving efficiency.

Self-Service Networking with VPCs and Multi-Tenancy

VPCs provide isolated network environments within a shared infrastructure. You can use Virtual Private Cloud (VPC) constructs to provision and manage network resources to business units or project teams independently, with granular control over networking policies and resource allocation.

By creating projects, you can isolate security and networking objects across tenants in a single NSX deployment. You enable multi-tenant deployment and consumption of NSX services for the provider administrator and the tenants or organizations. The provider administrator can create projects, assign users and groups, and allocate quotas to restrict the configurations available to the tenants.

NSX Documentation

For more information about the built-in NSX capabilities, see the NSX documentation.

VCF Operations Overview

VMware Cloud Foundation Operations® (formerly VMware Aria Operations) or VCF Operations helps your organization build, manage, operate, and secure your private cloud infrastructure by deploying and maintaining its fleet-level components, providing visibility and enhanced performance across the workload and infrastructure stack, and helping your organization stay compliant with regulatory standards and organizational guidelines.

VCF Operations has the following main functional areas, reflecting the main categories of operations in a VCF private cloud:

- Build Your Private Cloud with VCF Operations
- Manage Your Private Cloud with VCF Operations
- Operate Your Private Cloud in VCF Operations
- Protect Your Private Cloud with VCF Operations

Build Your Private Cloud with VCF Operations

By using VCF Operations, you can add infrastructure to your private cloud as workload domains or standalone vCenter instances and can manage the lifecycle of your private cloud by providing the ability to patch and upgrade your infrastructure.

VCF Operations is also the central hub for managing software binaries using software depots . VCF Operations also enables extensibility by providing developers with access to APIs and SDKs through its developer center area.

Workload Domain Creation

You can create workload domains or import vCenter instances as workload domains in a VCF Instance from the VCF Operations inventory. Accordingly, VCF Operations deploys a vCenter instance, an NSX Manager instance (which can be shared with other workload domains) and adds a default vSphere cluster with ESX hosts to vCenter and NSX Manager, optionally with enabled Supervisor functionality.

For more information, see the Managing VCF Domains in VMware Cloud Foundation documentation.

For information about VCF logical constructs, see VCF Taxonomy.

Lifecycle Management and Software Depot

The fleet lifecycle component provides a unified framework to upgrade, patch, back up, restore, and maintain VCF Operations, VCF management services, and VCF Automation. You can also deploy additional components after the initial VCF deployment or convergence. Similarly, you use SDDC lifecycle management to perform lifecycle management on the components running on VCF Instances. Component binaries are stored in one or more software depots - one at the fleet level and optionally one or more software depots at the instance level.

For more information, see the Upgrading to VMware Cloud Foundation 9.1.x, Lifecycle Management, Adding or Removing VCF Components Post Deployment, and Component Backup and Restore of VMware Cloud Foundation documentation.

Manage Your Private Cloud with VCF Operations

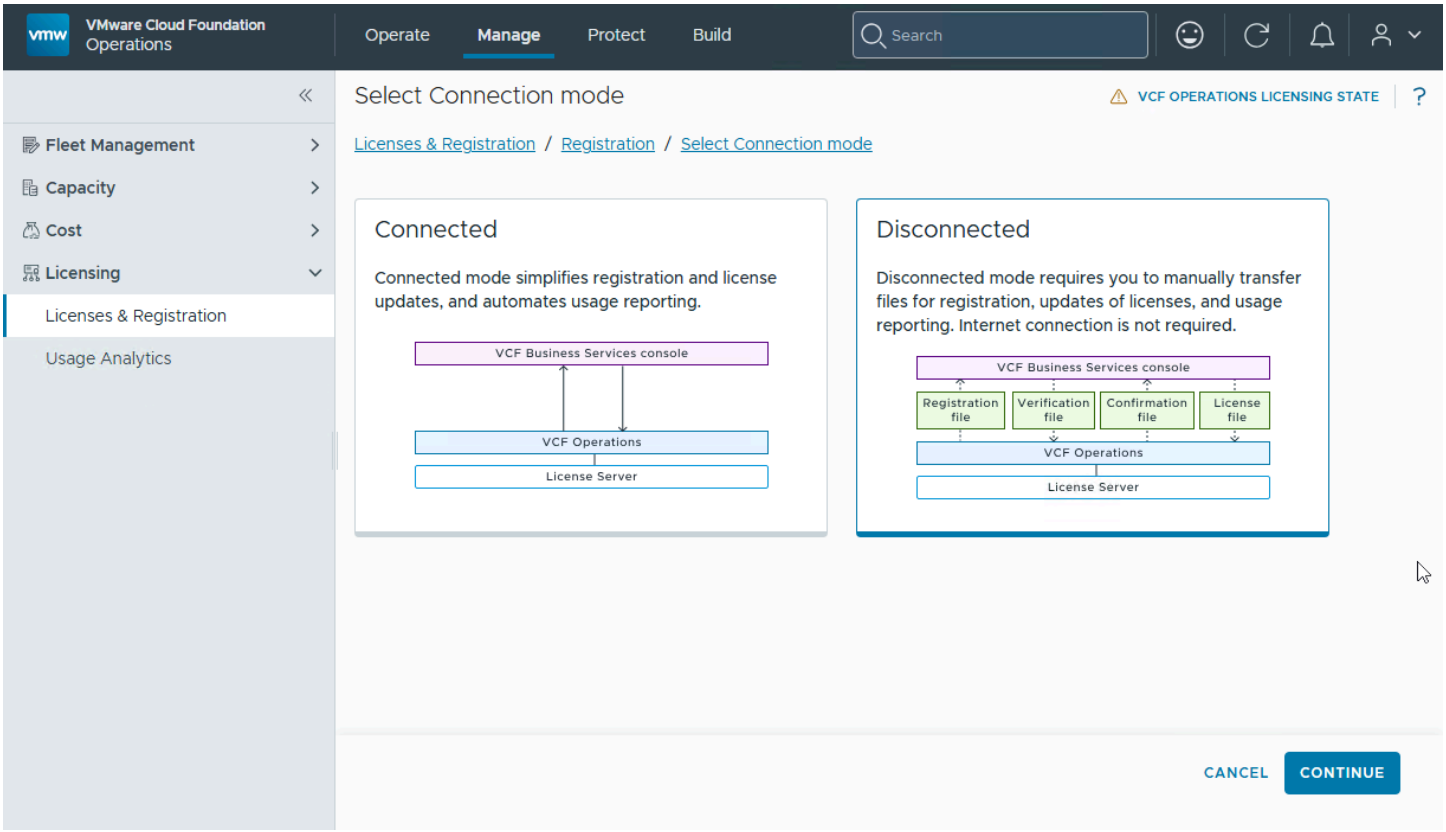
In VCF Operations, you can make your infrastructure compliant and standardized using automated updates across all workload domains, including identity-based access, license entitlement, and fleet-wide certificate and password maintenance.

Licensing

With a unified view in VCF Operations, you can easily manage licensing for your VCF and vSphere Foundation infrastructure. VCF Operations provides a comprehensive overview of VCF license allocation and usage in a private cloud. VCF Operations also provides a view of the older licenses that are pre-version 9.0.

VCF provides license management in two modes - for connected environments and disconnected environments. You can upgrade your licenses and you can choose when and how to monitor VCF assets across the entire VCF stack.

Figure 2: Licensing in VCF Operations



For more details on how to use license management, see the Licensing documentation.

Fleet Management

By using the fleet management functionality of VCF Operations, such as identity and access management, management of certificates and passwords, you can centrally manage your VCF components.

Figure 3: Fleet Management in VCF Operations

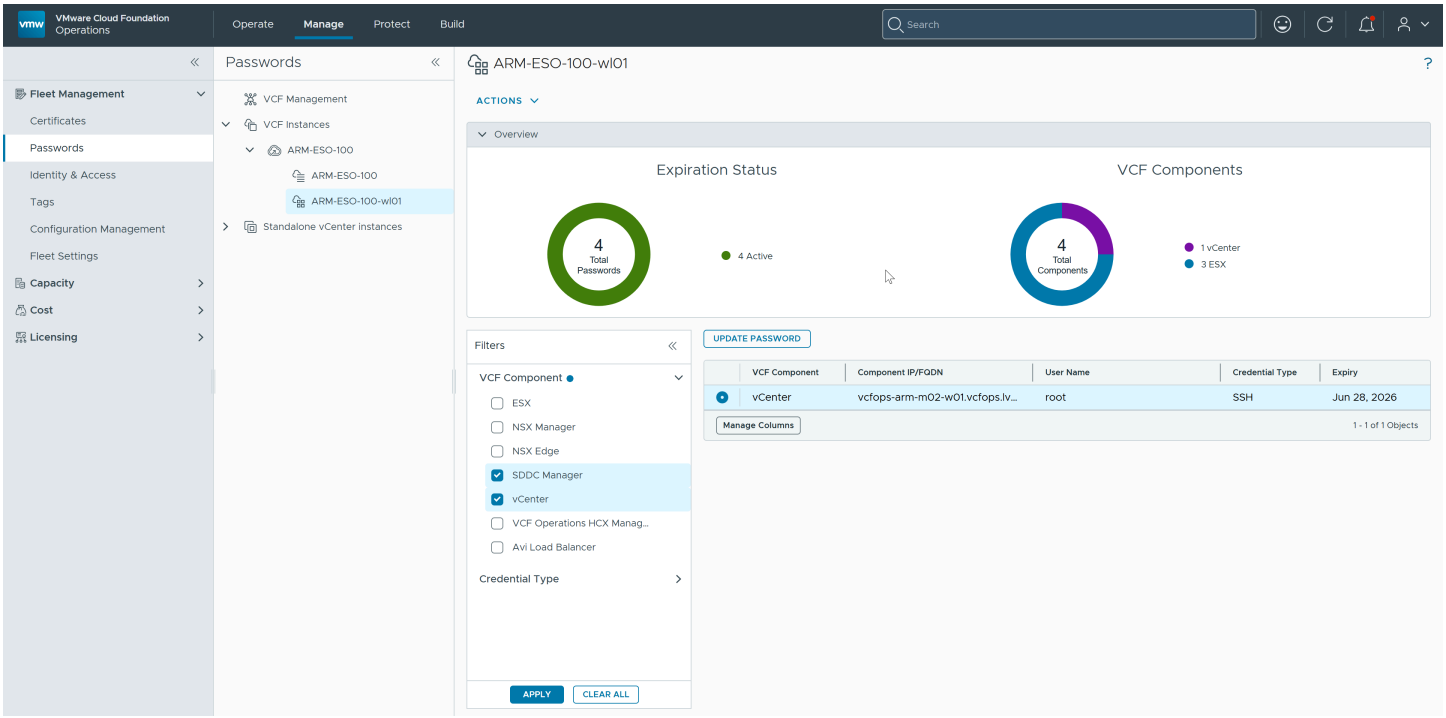


Table 1: Fleet Management Capabilities in VCF Operations

Fleet Management Capability	Description
Password Management	Unified password management in VCF centralizes account password management for most the components of the VCF private cloud. It offers a consolidated view of accounts, their statuses, and remediation needs. You can easily identify expiring or soon-to-expire passwords. For more details on how to use password management, see the Managing Passwords for VMware Cloud Foundation Components documentation.
Certificate Management	VCF Operations introduces a unified, non-disruptive TLS certificate management system, allowing centralized control of certificates across most of the components in the VCF private cloud. The system provides automated workflows, integration with various certificate authorities, and auto-renewal capabilities. Features include viewing certificates and alerts, auto-renewal for VMCA, MSCA, and OpenSSL CA, replacing default certificates with enterprise CA certificates, CA configuration and CSR generation, and certificate expiry notifications. For more details on how to use certificate management, see the Managing Certificates in VMware Cloud Foundation documentation
Identity and Access Management	You can use single-source configuration in VCF Operations for identity and access management. In this way, you can use the same identity configuration across VCF components and across multiple geolocations, offering Single Sign-On (SSO) and centralized management of federated access, enhancing both flexibility and user experience. For simplified access management of VCF users, you can use VCF roles across VCF components, such as vCenter, NSX, VCF Operations, VCF Automation, and other. You can assign predefined VCF roles for functions such as admin, viewer, and other. For more details on how to use identity and access management, see the Managing Identity and Access With VCF Single Sign-On documentation.
Tag Management	By using unified tag management, you can manage tags across vCenter instances from a central location. By using the tag management capabilities you can perform the following tasks: • Create, edit, or delete categories and tags from a single pane of glass. • Unify existing tags across vCenter instances, assign and view tag assignments, and define consistent tagging practices at scale. • Import tags and categories from converged or upgraded environments. For legacy VCF Operations (formerly VMware Aria Operations) users, legacy tags continue to be available independently. • Create tags and categories. • Push tags and categories • Assign tags to vCenter instances. For more details on how to use tag management, see the Tags and Categories Overview documentation.
Configuration Management	Unified configuration management provides features like scheduled drift detection for vCenter and cluster objects through VCF Automation or VCF Operations orchestrator., enabling proactive detection of configuration deviations from the desired state. You can manage global templates across vCenter instances, integrate with Git, and schedule drift detection through VCF Automation or VCF Operations orchestrator. For more details on how to use configuration management, see the Managing the Configuration of your Environment documentation.

Fleet Management Capability	Description
Fleet Settings	You can configure common settings for the appliances that are running the VCF fleet components, such as password policy and DNS and NTP settings. For more details on DNS and NTP configuration, see Managing Infrastructure Services for VMware Cloud Foundation. Password policy enables you to be compliant with enterprise security standards for managing local accounts. Your organization might have specific requirements defined either by the organization itself or through an industry compliance standard that prescribes the changes that you must make to the default policy configuration. After you configure the password policy, you can use VCF Operations to rotate or manually update the passwords of the VCF components. For more details on password policy configuration, see Create a Password Policy for VMware Cloud Foundation Componentsand Apply a Password Policy for VMware Cloud Foundation Components.

FinOps and Capacity Management

By using capacity management in VCF Operations, you can assess, optimize, and plan capacity across your on-premises environment. Cost management helps you analyze your infrastructure expenses. In this way, you can identify areas that are most suitable to use capacity management for optimal utilization of your resources.

Figure 4: FinOps and Capacity Management in VCF Operations

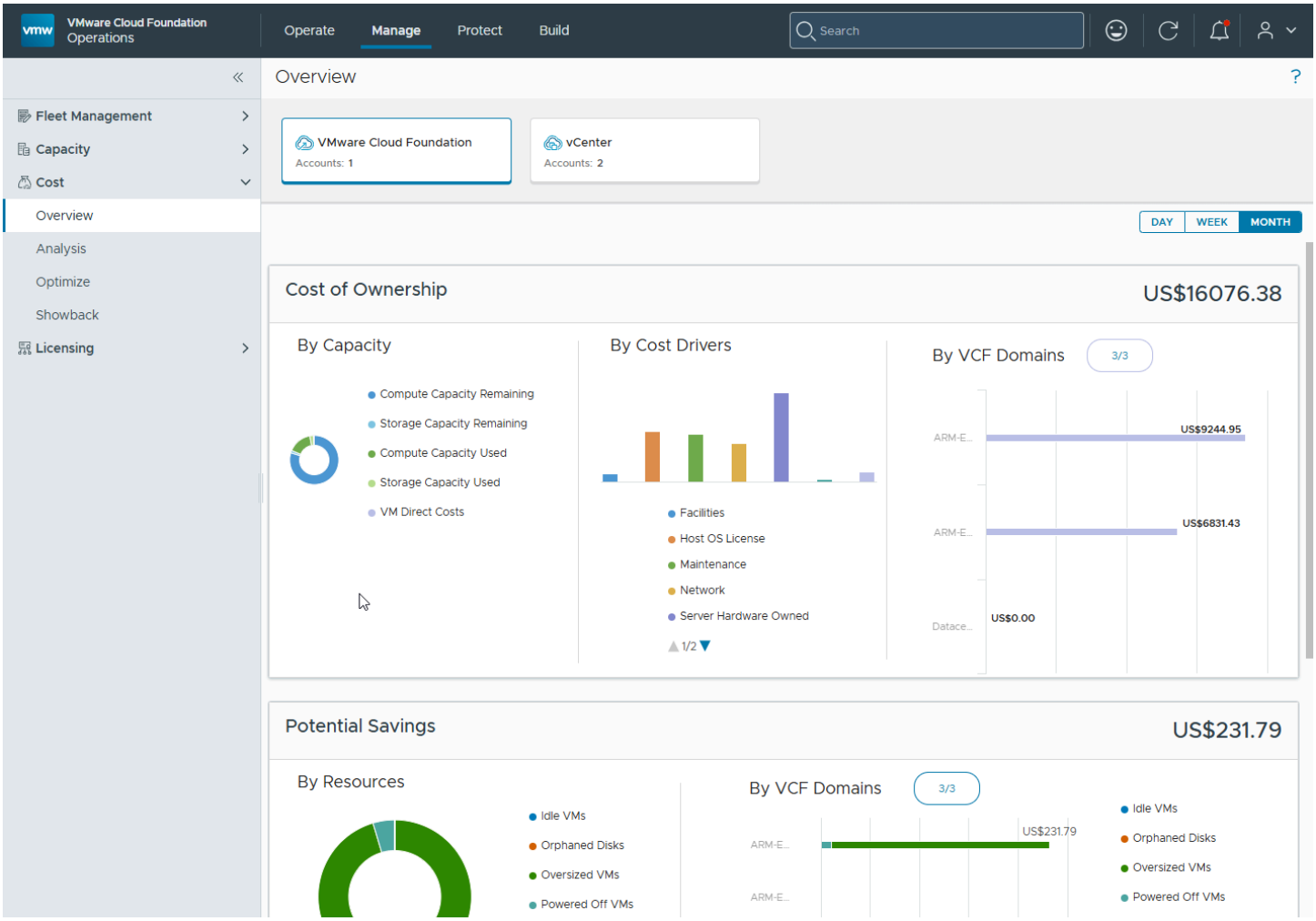


Table 2: FinOps and Capacity Management Capabilities in VCF Operations

Resource Optimization Capability	Description
Capacity Management	Capacity management consists of 3 sets of capabilities - assess, optimize, and plan. Use capacity planning to evaluate the capacity requirements of your organization based on historical resource utilization and real-time predictive projections to plan ahead. Capacity assessment lets you evaluate the available and consumed capacity (compute and storage). For more information, see the Viewing Object Capacity in the Capacity Tab, What-If Analysis - Workload Planning: Traditional and What-If Analysis: Modeling Workload, Capacity, or Migration Planning documentation.
Cost Management	You can use the cost management capability in FinOps to monitor infrastructure-related costs and associated services or license expenses with greater precision. It also enables showback and chargeback capabilities, helping application teams understand and manage the cost of the infrastructure they consume. VCF Operations also supports costing for vSphere Kubernetes Service (VKS) cluster. VCF Operations calculates the cost for VKS cluster, Supervisor cluster, and Kubernetes node, by aggregating the cost of the connected VMs. For more information, see the Configuring Cost documentation.

Operate Your Private Cloud in VCF Operations

The comprehensive observability hub in VCF Operations, delivers a single pane of glass for monitoring operational health and performance. To enable proactive troubleshooting and quick root-cause analysis across the entire infrastructure, it synthesizes critical information, including system metrics, logs, and network traffic flows. Additionally, it provides environmental analytics through sustainability scores, allowing organizations to align infrastructure performance with energy-efficiency goals.

Core Monitoring

The Inventory of VCF Operations is a collection of all objects in your VCF fleet that are discovered and monitored by VCF Operations. The inventory includes objects in all VCF Instances in your environment, including management domains and workload domains, and all standalone vCenter instances. The inventory is the primary entry point for observability and health assessment. It provides real-time visibility into object health across the VCF fleet to surface active alerts, performance bottlenecks.

Figure 5: Examining the Health of the Objects in a vSphere Cluster

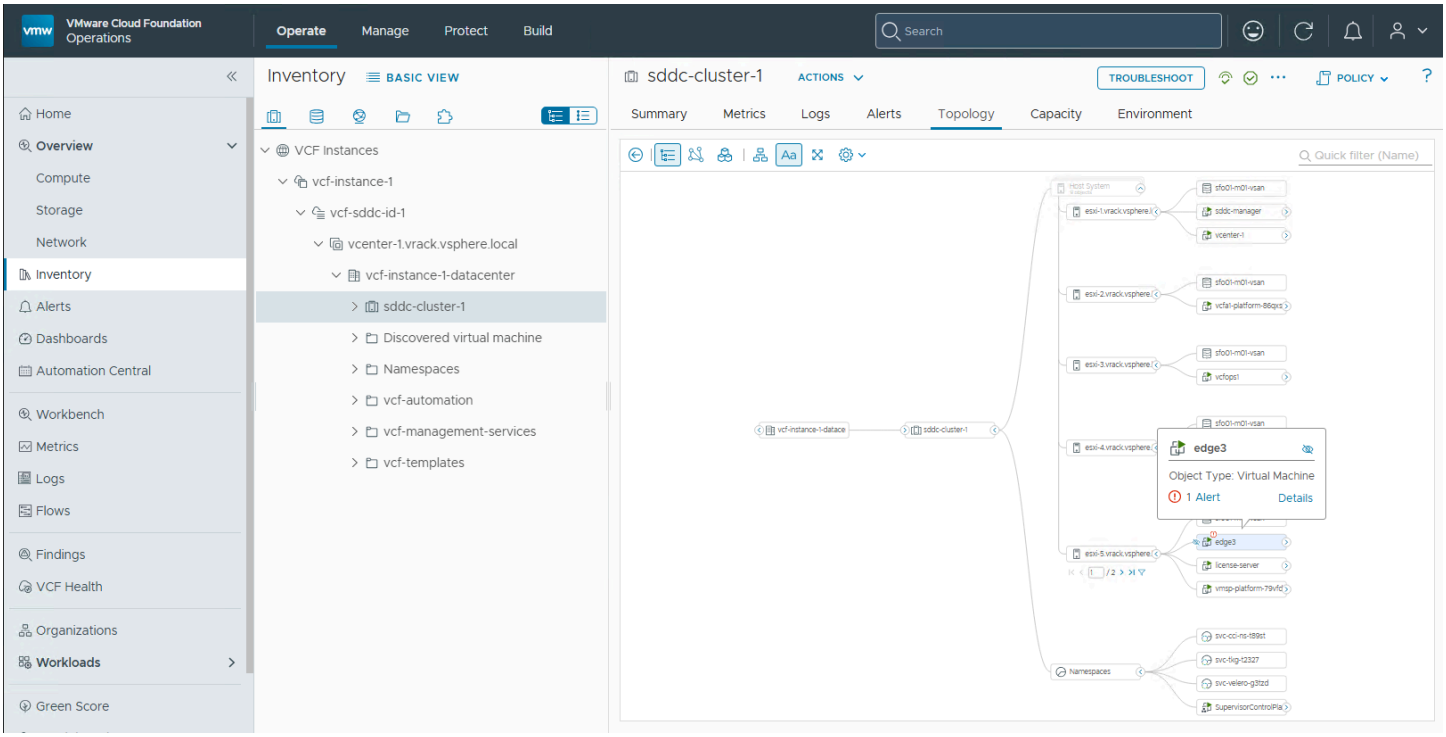
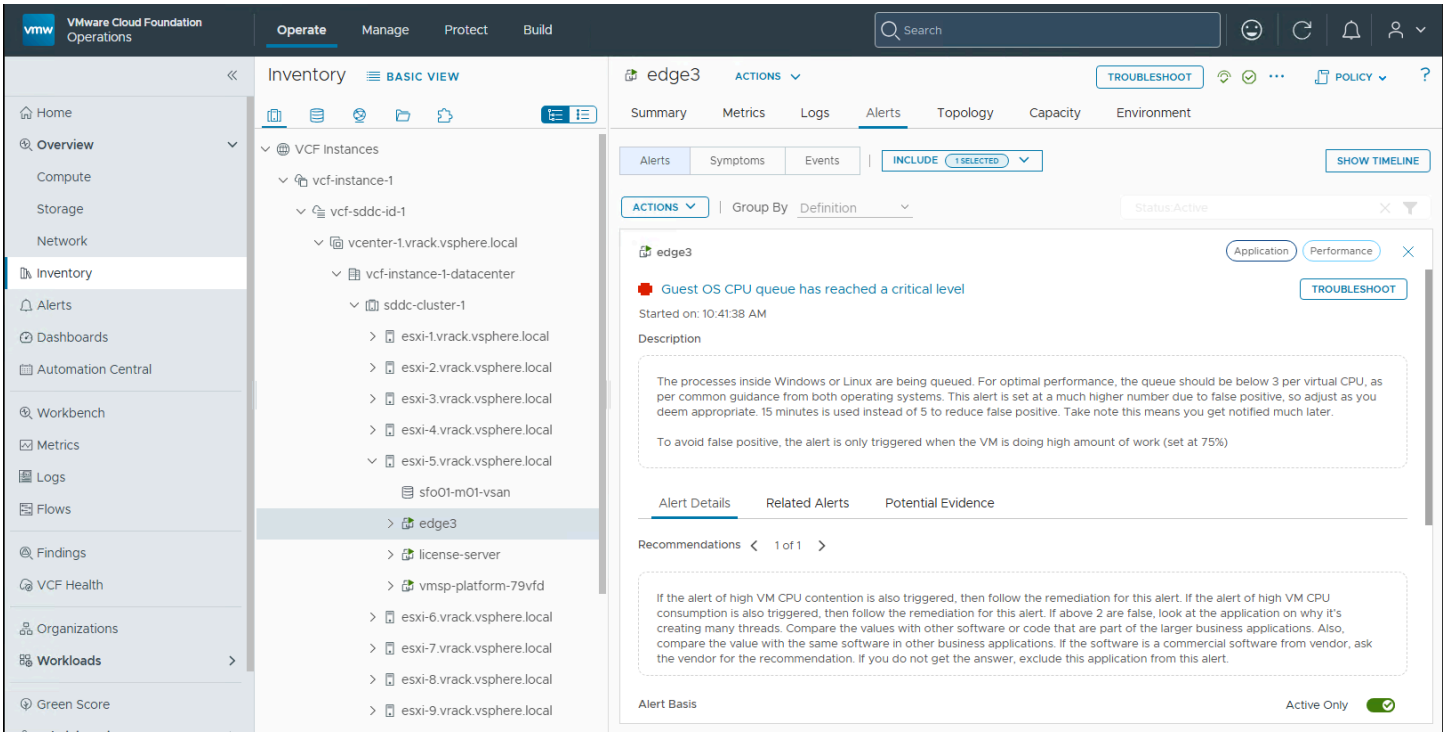


Figure 6: Locating Alerts for an Object in the Inventory



Diagnostics and Analysis

This functional area integrates disparate data types, including structured metrics, unstructured logs, and network flow data, into unified workflows. Instead of isolated monitoring, these features prioritize data correlation to isolate the root cause of performance degradations or system failures across the management and workload domains.

Diagnostics and Analysis Capability	Description
Workbench	Provides teams with the ability to collaborate, build a knowledge base of issue investigations which can be saved for future reference, and can enhance troubleshooting by viewing real-time data. For more details on Workbench, see the Troubleshooting with the Workbenchdocumentation.
VCF Health	Continuously monitors the operational state of VCF components in your environment to ensure they are functioning as expected. By proactively identifying potential issues such as expired certificates, NTP drifts, DNS misconfigurations, and configuration errors, you can take preventative actions before these issues escalate into significant problems. By using proactive monitoring, you can maintain a stable, secure, and efficient environment, reducing the risk of downtime and operational disruptions. For more information on how to use VCF Health, see the Monitor Your Environment and Prevent Issues with VCF Health documentation.
Diagnostics	Automatically detects known issues and critical vulnerabilities within the VCF software stack, providing detailed descriptions, root cause insights, and actionable remediation steps to resolve problems and mitigate risks. The Diagnostic Findings feature offers a consolidated view of known product issues, VMSA-based security exposures, and best-practice recommendations across your VCF environment. Diagnostic Findings helps VCF administrators gain visibility into potential issues and risks, prioritize resolution efforts, and achieve uninterrupted operations. For more information how to use Diagnostics Findings, see the What are Diagnostic Findings documentation.
Log management	Provides deep dive into infrastructure logs, with query and filtering options to troubleshoot issues across your virtualized environment. You can analyze and visualize log data, and create custom queries for ongoing monitoring. For more information on how to perform log analysis, see the Configuring and Analyzing Logs documentation.
Log Assist (Transfer Support Bundle)	Streamlines the process of manually gathering and uploading support bundles used by VMware Technical Support Engineers (TSEs) to help solve customer technical issues. With the Log Assist feature, you can transfer support bundles to Broadcom Support from VCF Operations for the purpose of troubleshooting an open Support Case. For more information about how to use Log Assist, see the Use Log Assist to Accelerate Your Support Cases documentation.
Storage	View and manage centrally the operational state of your VCF storage, reviewing storage capacity and performance. Storage Operations provides the following capabilities: • Centralized view of storage inventory, capacity utilization, and key performance statistics • Visibility into the health of vSAN clusters, and drill down experience for further analysis • Visibility into vSAN deduplication and compression for both vSAN ESA and OSA deployments • vSAN Performance Diagnostics For more information about how to monitor your storage, see the Monitoring Storage Operations documentation.
Flows	Provides the option to view and manage the operational state of your VCF networking, allowing for streamlined troubleshooting and ensuring network performance is optimized. For more information on how to monitor network operations, see the Network Operations documentation.

Diagnostics and Analysis Capability	Description
Integrations	Provides integration of VCF Operations with VCF components and other second- and third-party systems by using management packs. You can install management packs that are already included in the VCF Operations deployment or download management packs from the VCF Solutions Catalog. See the Installing and Configuring Management Packs in VMware Cloud Foundation Operations from the Solutions Catalog documentation. You can also create a custom management pack in VCF Operations. For more information, see the Management Pack Builder documentation.

Sustainability

Green Score is used to measure sustainability in digital operations through data center virtualization. Green Score consists of multiple parameters for effective measurement of the score. Virtualization plays a crucial role in optimizing resource utilization, helping reduce hardware footprints in data centers and further enabling savings in power consumption and floor space. This improves the overall efficiency and curbs carbon emissions resulting from IT infrastructure and operations.

For more information, see the Configuring Green Score documentation.

Workload Operations

Workload operations ensure that critical applications are running properly by monitoring their performance, availability, and end-user experience. For container-based applications, VCF Operations provides native integration with vSphere Supervisor which enables automatic monitoring of Supervisor instances and vSphere Kubernetes Service (VKS) clusters at a granular level, shortening troubleshooting time.

A business application consists of a set of interconnected applications or services running on one or more virtual machines which together offer a service to users.

For more information, see the Workload Monitoring and Observability documentation.

Protect Your Private Cloud with VCF Operations

VCF Operations provides continuous, real-time assessment of the infrastructure's security posture by monitoring encryption readiness and critical security advisories. This defensive stance is extended by robust auditing and compliance validation against industry benchmarks.

Security Operations

The Security Operations dashboard is designed to provide a comprehensive view of your security posture in VCF. This dashboard consolidates elements that are related to user and infrastructure security to help you identify areas that might require attention and tightening based on your security exposure. The dashboard covers host encryption, host mode compliance, vSAN cluster encryption, advisories for CVE violations, certificate health, and VM encryption status.

For more information on how to use security operations, see the Monitoring Security Operations documentation.

Governance and Resilience

Governance and Resilience Capability	Description
Audit Records	Each audit record has information which helps you analyze the user actions in your VCF environment. In VCF Operations, you can examine the diverse audit records that are generated across the supported VCF components from a central location. For more information, see the Audit Records and Audit Trail documentation.
Compliance	Ensure compliance for your VCF infrastructure and cloud environments against optimum configurations, maintaining a consistent configuration across SDDC components, reducing organizational and business risk. For more information, see the Viewing and Configuring Compliancedocumentation.

Governance and Resilience Capability	Description
Protection and Recovery	The **Protection and Recovery** area of VCF Operations shows you information about Protection and recovery, vSphere Replication, vSAN snapshots, and replication (local and remote snapshots), VMware Live Recovery Cloud, and VMware Virtual Volume replication. Protection and recovery dashboards help you quickly understand whether the VMs in your environment are being protected and are recoverable in your on-premises and cloud environments. **Protection and Recovery** dashboards also show whether you have enough protected storage capacity to protect your cloud-based VMs. For more information, see Protection and Recovery Dashboards for VCF Operations in the *Protection and Recovery* documentation.

VCF Automation Overview

VMware Cloud Foundation Automation® (formerly VMware Aria Automation) or VCF Automation enables IT teams and cloud service providers (CSPs) to deliver a self-service private cloud for AI, Kubernetes, and VM-based applications. The solution simplifies the process of provisioning and scaling a multi-tenant private cloud with out-of-the-box Infrastructure as a Service (IaaS) offerings, accelerating the time to market for applications while maintaining control using policy-based governance.

VCF Automation has the following capabilities:

- Cloud Services. Provides a set of cloud services to meet the needs of application teams.
- Provider Management. Supports IT teams and cloud service providers to effectively manage, deliver, and scale cloud services across multiple vCenter and VCF instances.
- Organization Management. Provides capabilities that enable the organization administrator of a Line of Business or a tenant to further organize and govern resources allocated to them among application teams.
- vSphere Supervisor. Provides the capability to run Kubernetes workloads directly on ESX hosts and to create upstream Kubernetes clusters within dedicated namespaces.

Cloud Services Overview

VCF Automation provides a set of cloud services which are accessible on a self-service basis using multiple interfaces, including UI, CLI, and API.

With this interface, users can provision the following types of workloads:

- VMs
- Kubernetes workloads
- Networking
- Volumes
- Secret stores
- Databases
- Harbor container registries
- External DNS
- Certificates
- AI workloads

Additionally, application teams can also provision resources on a self-service basis from Catalogs which are backed by pre-configured workload templates. These self-service options help developers focus on application development instead of managing the underlying infrastructure.

For more information about cloud services, see the Building Cloud Applications documentation.

Provider Management Overview

IT teams and CSPs can model multi-tenancy by using the concept of Organizations in VCF Automation, effectively allocating resources and managing users specific to each Organization.

This capability provides isolated environments with dedicated network setup. IT Teams and CSPs can monitor resource usage, control permissions, and provide managed services while ensuring security through comprehensive network isolation.

For more details on provider management, see the following resources:

- Provider Management documentation. Explains how to use provider management.

Organization Management Overview

VCF Automation improves governance for users and resources by enabling organization administrators create and assign projects and vSphere Namespaces tailored for different application teams.

Organization administrators can set resource limits and policies while delegating network management through Virtual Private Clouds (VPCs) for application teams. Additionally, Policy as Code enables organization administrators to enforce instance-centric policies for Infrastructure as a Service (IaaS) resources from a central location for compliance and reducing the risk of human error.

For more details on organization management, see the following resources:

- Organization Management documentation. Explains how to use organization management.

vSphere Supervisor Overview

VMware vSphere Supervisor ® is the foundation of VCF consumption and can be enabled during workload domain creation or directly in vCenter.

It transforms vSphere clusters into modern application platforms by embedding a platform-level Kubernetes control plane directly into ESX hosts, exposing a unified declarative API surface. This integration allows end users to provision and manage workloads, including VMs, containers using vSphere Pods, full Kubernetes clusters through VKS, OCI image registries, and data services through a consistent API interface.

vSphere Supervisor maintains enterprise-grade governance, security, and operational consistency through vCenter, VCF Operations, and VCF Automation. Its vSphere Namespaces provide the foundation for multi-tenancy, enabling GitOps-based workflows, and delivering cloud-like self-service in private data centers.

For an overview of vSphere Supervisor concepts and how they apply in VCF Automation, see the vSphere Supervisor Platform and Building Cloud Applications documentation.

What Is vSphere Foundation?

VMware vSphere Foundation (vSphere Foundation) is an enterprise-grade workload platform for modern infrastructure delivering virtualized benefits, simplified management, cost efficiency, scalability and serves as the core foundation for VCF.

vSphere Foundation provides the following capabilities:

- Intelligent operations management for visibility and infrastructure optimization.
- Hyperconverged infrastructure integrating compute and storage virtualization for resource management.
- Simplified deployment and scalability enabling faster application delivery and future ready infrastructure.
- Unified service-oriented platform, VCF management services, for centralized lifecycle management and operational functionality.
  In a vSphere Foundation environment, VCF management services provide lifecycle management of VCF Operations and license server, and log analysis.
- Unified workload platform for running consumer VMs and containers side by side in a native Kubernetes runtime.

vSphere Foundation does not include the cloud management and integrated automation capabilities that are available in VMware Cloud Foundation. See What Is VMware Cloud Foundation?.

For a feature comparison and differences between VMware Cloud Foundation and VMware vSphere Foundation, see Feature Comparison & Upgrade Paths on vmware.com.

See the following documentation:

- vSphere documentation
- VCF Operations Overview
- Constructs in VCF Management Services
- vSphere Supervisor Overview
- Components in VCF and vSphere Foundation

VCF Taxonomy

The VCF fleet, VCF Instance, and VCF domain logical constructs provide a modular approach to building and managing your private cloud. Each construct has a clear scope and components.

For information on designing a VCF platform according to industry recommendations and best practices, see the VCF Design documentation. For information about the components of a VCF deployment, see Components in VCF and vSphere Foundation.

Figure 7: VCF Fleet, VCF Instances, and VCF Domains

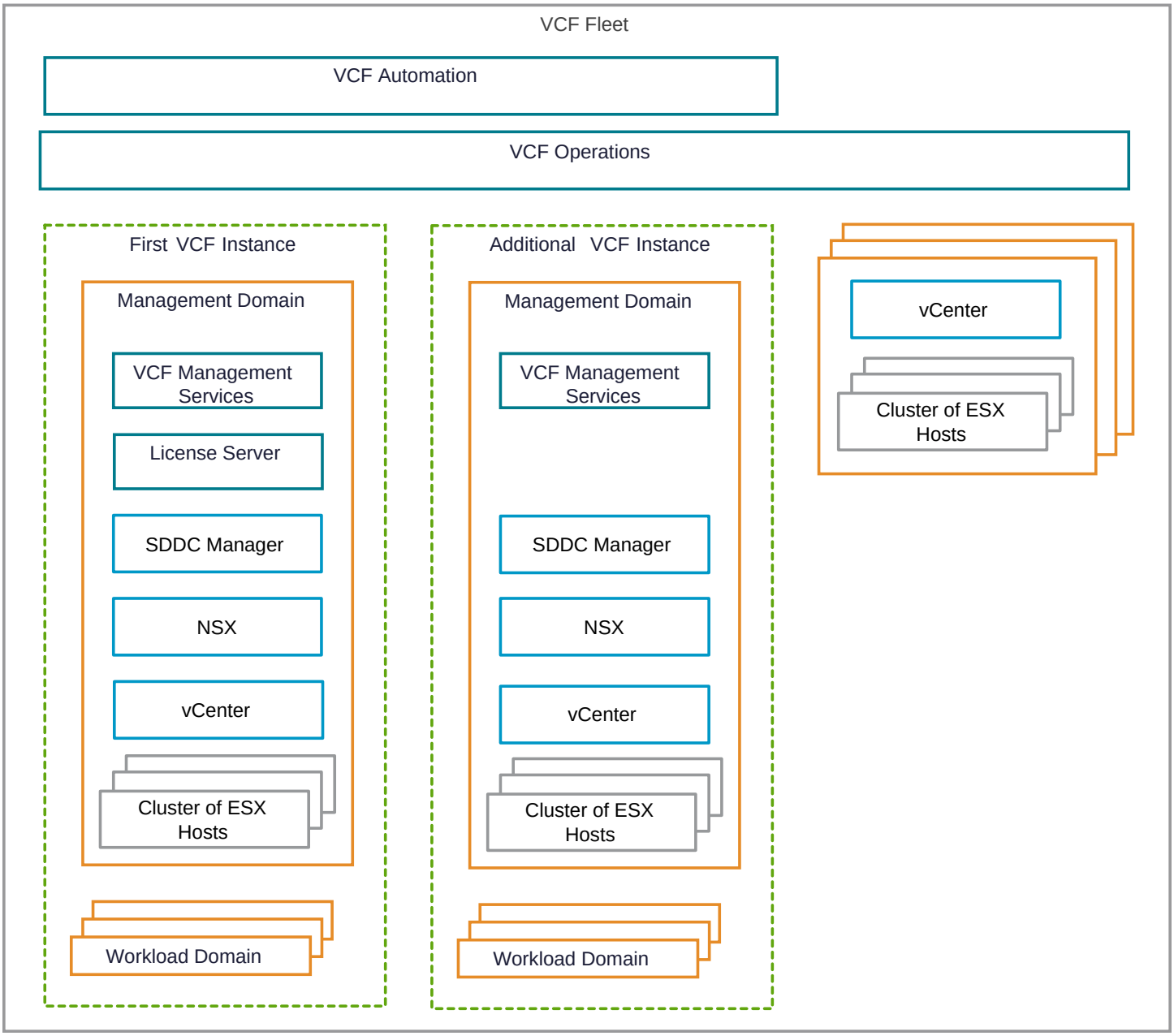


Table 3: Constructs of a VCF Private Cloud

Construct	Description	Components
VCF Instance	Compute, storage, networking virtual infrastructure that runs business workloads. A VCF Instance consists of VCF domains - a management domain and, optionally, workload domains. In a clean VCF deployment, the first vSphere cluster of a management domain runs the management components for the VCF Instance. In an environment where multiple vCenter instances have been converged to VCF, the management components can also be in vSphere clusters in the workload domains.	• SDDC Manager • NSX • vCenter • ESX • VCF management services • License server
VCF fleet	An environment that is managed by a single set of fleet-level management components - VCF Operations and VCF Automation. A VCF fleet contains one or more VCF Instances and might contain one or more standalone vCenter instances, managed by the VCF Operations instance for the fleet. The components of the VCF fleet are running in the management domain of the first VCF Instance.	• One or more VCF Instances • One or more standalone vCenter instances • VCF Operations • VCF Automation
VCF private cloud	The highest level of management and consumption for the underlying software-defined data center resources. A VCF private cloud can contain one or more VCF fleets.	• One or more VCF fleets

VCF Domains

A VCF domain in a VCF Instance consists of the following components:

- One vCenter instance.
- One or more vSphere clusters with vSphere HA and vSphere DRS enabled.
- At least one vSphere Distributed Switches per cluster for system traffic and NSX segments for workloads.
- One NSX Manager instance, either specifically for the VCF domain or shared between VCF domains, for configuring and implementing software-defined networking.
- Optional. One or more NSX Edge or Virtual Network Appliance clusters, added after you create the workload domain, that connects the workloads in the workload domain for logical switching, logical dynamic routing, and load balancing.
- One or more shared storage allocations as principal or supplemental storage.

Table 4: Types of VCF Domains

VCF Domain Type	Description
Management domain	The management domain is created during the deployment or convergence process by the VCF Installer. The management domain contains the components of the VCF Instance and, in the first VCF Instance, License server, VCF Operations, VCF management services, and VCF Automation. In addition to the components that are common for each VCF domain, the management domain includes the SDDC Manager appliance that adds workflows for infrastructure management automation to the VCF Operations instance for the VCF fleet, and VCF management services that are working on the instance level.
Workload domain	You create workload domains in VCF Operations to run consumer workloads. For the first workload domain in your environment, in the management domain, VCF Operations deploys a vCenter instance , and if required by your organization, an NSX Manager instance. For each additional workload domain, VCF Operations deploys an additional vCenter instance. New workload domains can share the same NSX Manager cluster with an existing workload domain or with the management domain, or you can deploy a new NSX Manager cluster.

Constructs in VCF Management Services

VCF 9.1 introduces VCF management services as a unified service-oriented platform for centralized lifecycle management and operational functionality.

Construct	Description	Available In
VCF services runtime	A common architectural runtime that provides a shared foundation for VCF services across the VCF fleet and VCF Instances. The VCF services runtime instance for VCF management services has the following node architecture: • VCF services runtime for VCF management services represents a Kubernetes cluster with the following node VMs: – Control plane VMs run the Kubernetes control plane. – Multiple worker node VMs run the VCF management services components, such as, fleet lifecycle, identity broker, log management, software depot, Salt RaaS, Salt master, and real-time metrics, and telemetry. • Number of nodes according to the VCF deployment model. – Simple. A single control plane node VM and three worker node VMs in the default management vSphere cluster. – High-availability. Three control plane VMs and a corresponding number of worker node VMs are distributed across ESX hosts of the default management vSphere cluster by using anti-affinity rule for high availability. • Component services with multiple replicas, such as identity broker, are distributed across worker nodes for resilience. • Components without high availability support are scheduled with reserved overhead so that they can restart on a running worker node if one fails. • Node self-healing re-creates failed or unhealthy nodes automatically, and component services rebalance once the new node is healthy. • The architecture is the same for fleet-level VCF services runtime clusters in the first VCF Instance and instance-level clusters in additional VCF Instances. The VCF services runtime instance for VCF Automation has the following node architecture: • VCF services runtime for VCF Automation represents a Kubernetes cluster of node VMs with combined control plane and worker node capabilities. • Number of nodes according to the VCF deployment model – Simple. A single node VM in the default management vSphere cluster. – High-availability. Three node VMs are distributed across ESX hosts of the default management vSphere cluster by using anti-affinity rule for high availability.	• VCF management services • VCF Automation
VCF management services	A modular suite of components that provide unified fleet-level and instance-local lifecycle, identity, and operational orchestration.	See Components in VCF and vSphere Foundation.

VCF Capabilities by Job Function

The VMware Cloud Foundation platform provides capabilities for all stages in the private cloud lifecycle and for all job functions working with the cloud's resources.

Job Functions Supported in a VMware Cloud Foundation Platform

The following functions are related to a VCF private cloud instance:

Table 5: Job Functions in VCF

Job Function	Description	
Virtual infrastructure administrator (VI administrator) or provider administrator	Build, manage, and operate cloud infrastructure to provide resources for running applications.	
Platform operator	Track and analyze performance and behavior of workloads.	
Organization administrator or tenant administrator	Enable consumers to operate autonomously and efficiently in the cloud resources allocated to the organization.	
Platform engineer	Manage, automate, and monitor cloud infrastructure for self-service workload provisioning.	
End users	Application developer	Develop and deploy VM- or container-based applications in the cloud
	DevOps engineer	Manage the Kubernetes and VM infrastructure on top of thevSphere Supervisor platform in VCF for continuous integration and continuous delivery of applications.
Roles related to special infrastructure	Storage administrator	Build and manage all centralized storage in the private cloud.
	Network administrator	Build and manage network services.

VCF Capabilities

You can access VCF capabilities that are specifically designed for your job function without the need to apply expertise outside of your knowledge domain.

Table 6: Capabilities for VI Administrators

Area	Capability
Build	• Deploy a new VCF platform • Converge existing virtual Infrastructure to a VCF platform • Upgrade to VCF 9.1
	• Deploy a workload domain • Import an existing vCenter instance as a workload domain • Add a vSphere cluster or an ESX host to a workload domain
	Configure a vSAN stretched cluster
	Manage network connectivity using transit gateways in vCenter backed by NSX Edge or Virtual Network Appliance clusters
	Manage networking using virtual private clouds in vCenter
	Manage virtual infrastructure in vCenter
	Manage the binaries required for the lifecycle management of VCF components
	Add VCF components after initial deployment
	Back up and restore the VCF fleet and its managed VCF Instances
	Manage the lifecycle of VCF components
Manage	Licensing
	Configure identity and access using VCF Single Sign-on
	Link vCenter instances for centralized inventory management

Area	Capability
	Manage certificates across the VCF fleet
	Manage passwords across the VCF fleet
	Manage tags across the VCF fleet
	Manage the configuration settings of vCenter instances and vSphere clusters
	Link vCenter instances for centralized inventory management
	Migrate workloads across workload domains and vCenter instances
Operate	Configure alerts
	Perform log analysis and troubleshooting
	Monitor network operations
	Monitor storage operations
	Perform infrastructure diagnostics and VCF health monitoring
	• Configure and use monitoring dashboards • Configure and use monitoring reports
	Perform workload monitoring and observability
	Manage cost and capacity
	Mitigate security risks and discover objects that are not compliant with a security standard
Consume	Deploy and configure vSphere Supervisor for running Kubernetes and VM workloads
	Manage infrastructure, and identity and access for consumer organizations

Table 7: Capabilities for Storage Administrators and Network Administrators

Area	Capabilities
Build	Manage vSAN storage
	Manage networking in NSX

Table 8: Capabilities for Platform Operators

Area	Capabilities
Build	Manage the binaries required for the lifecycle management of VCF components
	Back up and restore the VCF fleet and its managed VCF Instances
	Manage the lifecycle of VCF components
Manage	Link vCenter instances for centralized inventory management
	Manage certificates across the VCF fleet
	Manage passwords across the VCF fleet
	Manage tags across the VCF fleet
	Manage the configuration settings of vCenter instances and vSphere clusters
	Link vCenter instances for centralized inventory management
	Migrate workloads across workload domains and vCenter instances
Operate	Configure alerts
	Perform log analysis and troubleshooting
	Monitor network operations
	Monitor storage operations
	Perform infrastructure diagnostics and VCF health monitoring

Area	Capabilities
	• Configure and use monitoring dashboards • Configure and use monitoring reports
	Perform workload monitoring and observability
	Manage cost and capacity
	Mitigate security risks and discover objects that are not compliant with a security standard

Table 9: Capabilities for Organization Administrators and Platform Engineers

Area	Capabilities
Consume	Prepare the collection of VM-based workload blueprints and infrastructure services of your organization for end users
	Prepare the collection of VM- and container-based workload blueprints and infrastructure services of your organization for end users.

Table 10: Capabilities for DevOps Engineers and Application Developers

Area	Capabilities
Consume	Provision VMs
	Provision Kubernetes clusters
	Provision containers as vSphere Pods
	Provision persistent volumes
	Provision secrets
	Create databases using Data Services Manager
	Provision private AI workloads • Deep learning VMs • VKS Clusters with GPU • Generative AI applications

Getting Started with VMware Cloud Foundation

You start building a VCF platform by deploying a new platform, converging your existing virtual infrastructure to VCF, or upgrading existing your VCF platform. After you deploy your VCF Instances and VCF fleet, you can continue with adding workload domains and integrating them in VCF Automation for end-user consumption.

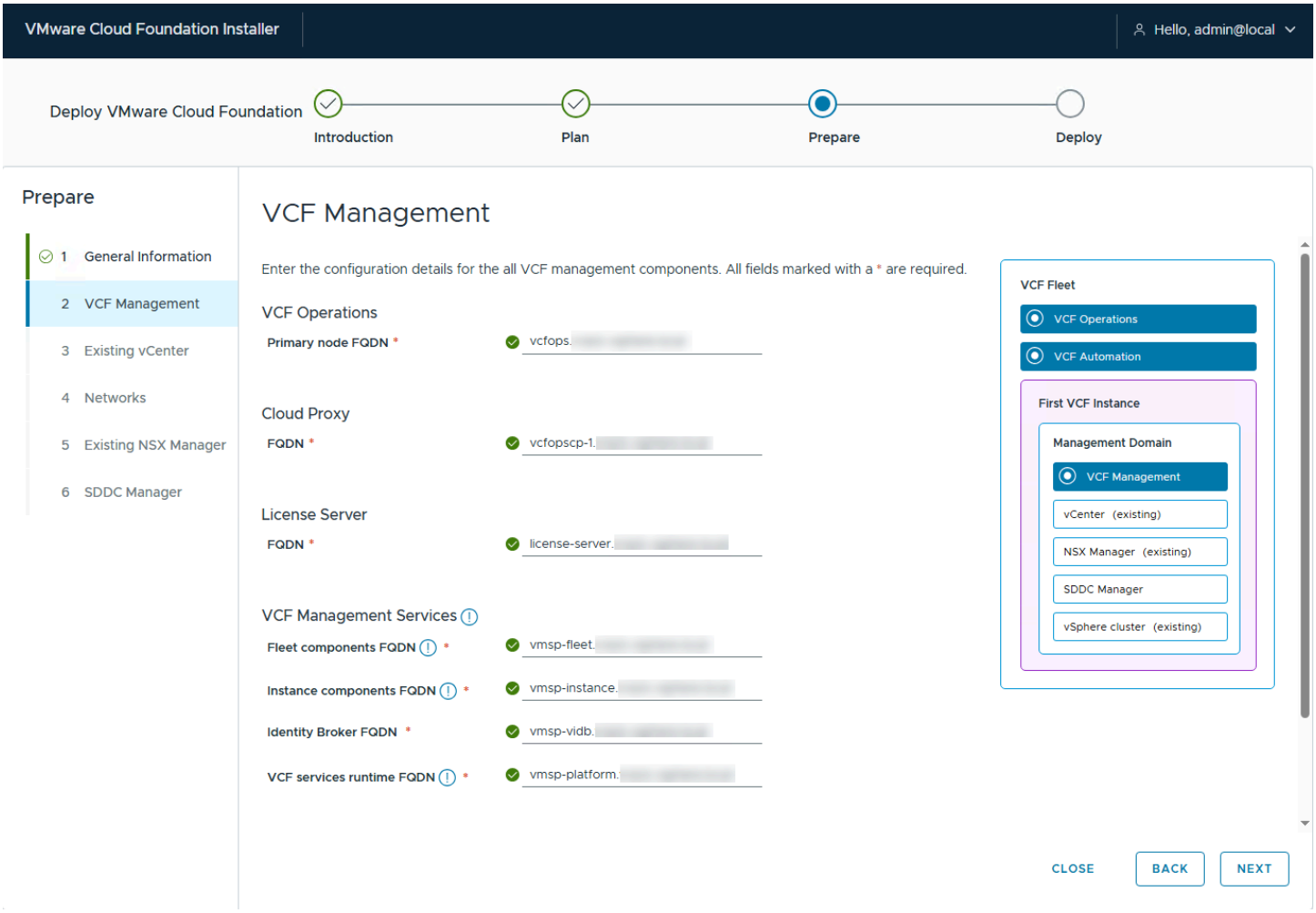
In this article:

- Deploy a VCF Fleet or a VCF Instance
- Add a vCenter Instance to the VCF Fleet
- Complete the Configuration of the VCF Fleet, VCF Instances, and Workload Domains
- Add Workload Domains to Your Cloud Organization Structure
- Design Blueprints
- List of VCF Capabilities
- What to Do Next

Deploy a VCF Fleet or a VCF Instance

The first step in building a VCF platform is deploying a VCF fleet or a VCF Instance in a VCF fleet.

Figure 8: Converging Your Environment to VCF 9.1 by Using the VCF Installer



Operation Type	Options

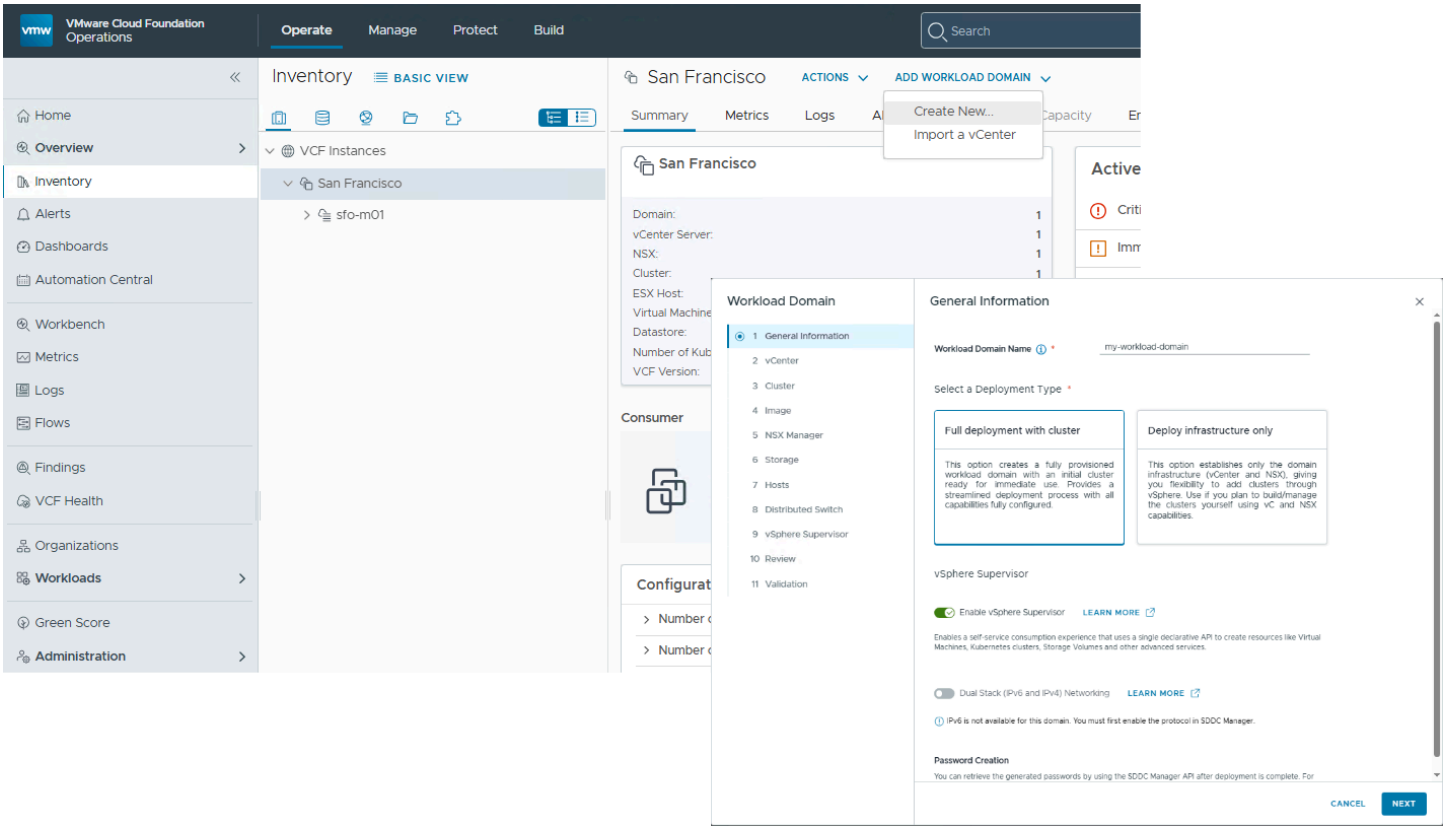
Operation Type	Options
Upgrade Your Management Domain to VCF.	• In a VCF 5.2.x environment, the high-level upgrade process is as follows: – Upgrade the existing VMware Aria components and SDDC Manager. Then, deploy VCF management services and the license server component. Upgrade the rest of the management domain components and deploy the rest of the VCF fleet components that you need. – If VMware Aria Operations is not part of your environment, deploy VCF Operations as a part of the upgrade operation. Then, deploy VCF management services and the license server component. Upgrade the rest of the VMware Aria and management domain components and deploy the rest of the VCF fleet components that you need. • In a VCF 9.0.x environment, the high-level upgrade process is as follows: Upgrade VCF Operations and SDDC Manager in the VCF Instances in your fleet. Then, deploy VCF management services and the license server component. Upgrade the rest of the fleet components and the management components of the VCF Instances.

Add Workload Domains to a VCF Instance

After you deploy the VCF fleet and VCF Instances, you can start adding workload domains with infrastructure for consumer workloads to the VCF Instances in one of the following ways:

- Deploy new workload domains.
  As part of workload domain deployment, you can enable vSphere Supervisor with NSX VPC on the initial cluster of the workload domain. You can then add the initial infrastructure of the workload domain to VCF Automation.
- Import vCenter instances, managing consumer workloads, as workload domains.
  You can import components that are compatible with VCF 5.x and upgrade to version 9.1 after you import them as a workload domain.

Figure 9: Adding Workload Domains in VCF Operations



Add a vCenter Instance to the VCF Fleet

You can add virtual infrastructure under a vSphere Foundation license to a VCF fleet. See vSphere as a data source in VCF Operations.

Complete the Configuration of the VCF Fleet, VCF Instances, and Workload Domains

After you deploy the VCF fleet or VCF Instance or add a workload domain to a VCF Instance, complete its configuration by performing the following tasks in VCF Operations or the vSphere Client. Assigning a license must be done right after you complete deployment.

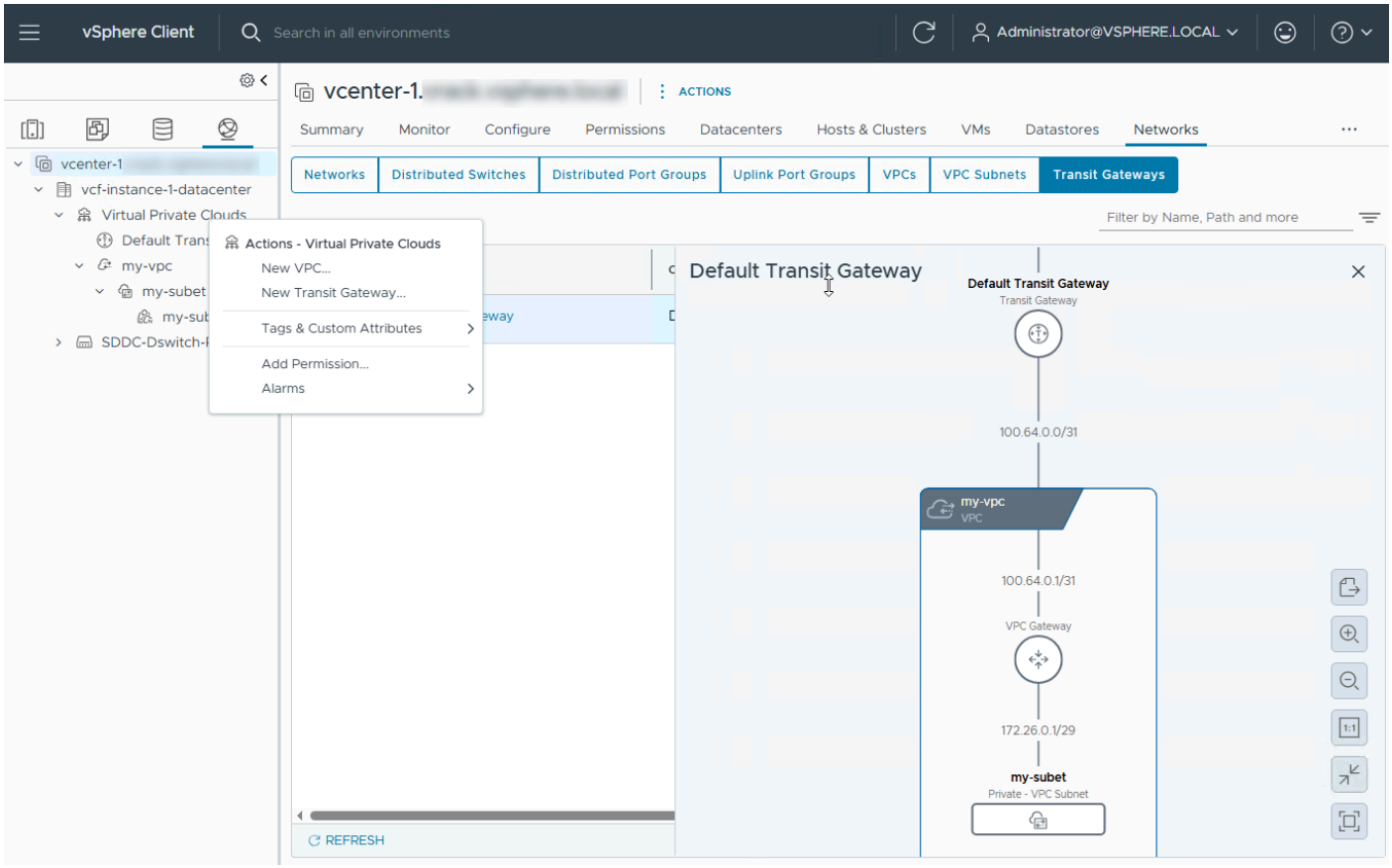
- Assign a license to the VCF fleet, VCF Instance or workload domain in VCF Operations.
- Configure software depot settings in VCF Operations.
- Configure identity and access management in VCF Operations.
- Configure backup of the management components in VCF Operations.
- Configure certificate management in VCF Operations and the vSphere Client.
- Configure password management in VCF Operations.
- Add clusters and individual ESX hosts to your VCF domains in the vSphere Client.

Add Workload Domains to Your Cloud Organization Structure

You can add workload domains to an organization in VCF Automation Provider Management for consumption by end users. You can create two types of organizations in VCF Automation - All Apps Organizations for workload domains with Supervisor clusters that run both containers and VMs, and a VM Apps Organization for workload domains with clusters that run only VMs.

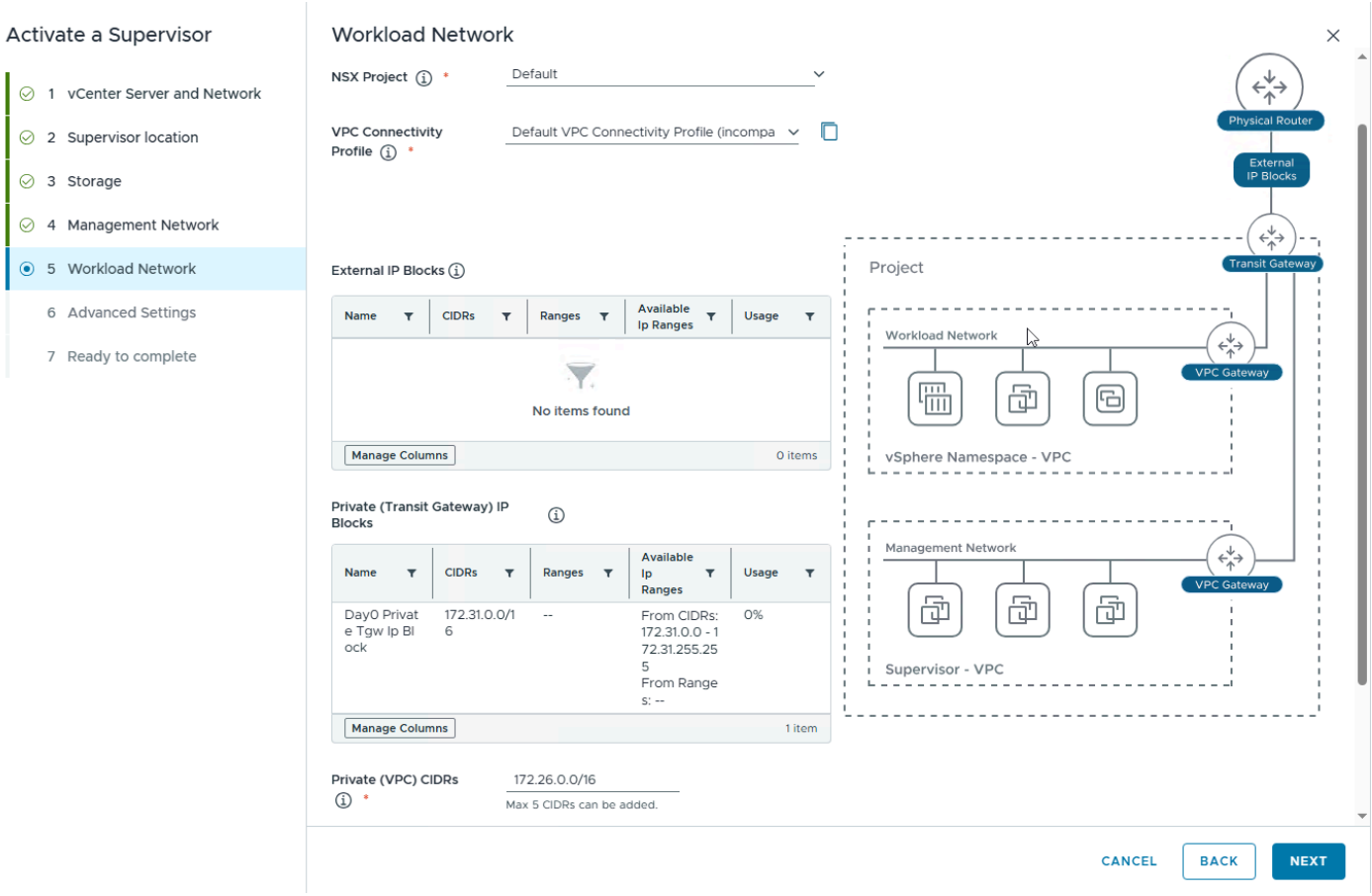
1. Set up network connectivity in the workload domains and create Virtual Private Cloud (VPCs) in vCenter.

Figure 10: Managing VPCs in the vSphere Client



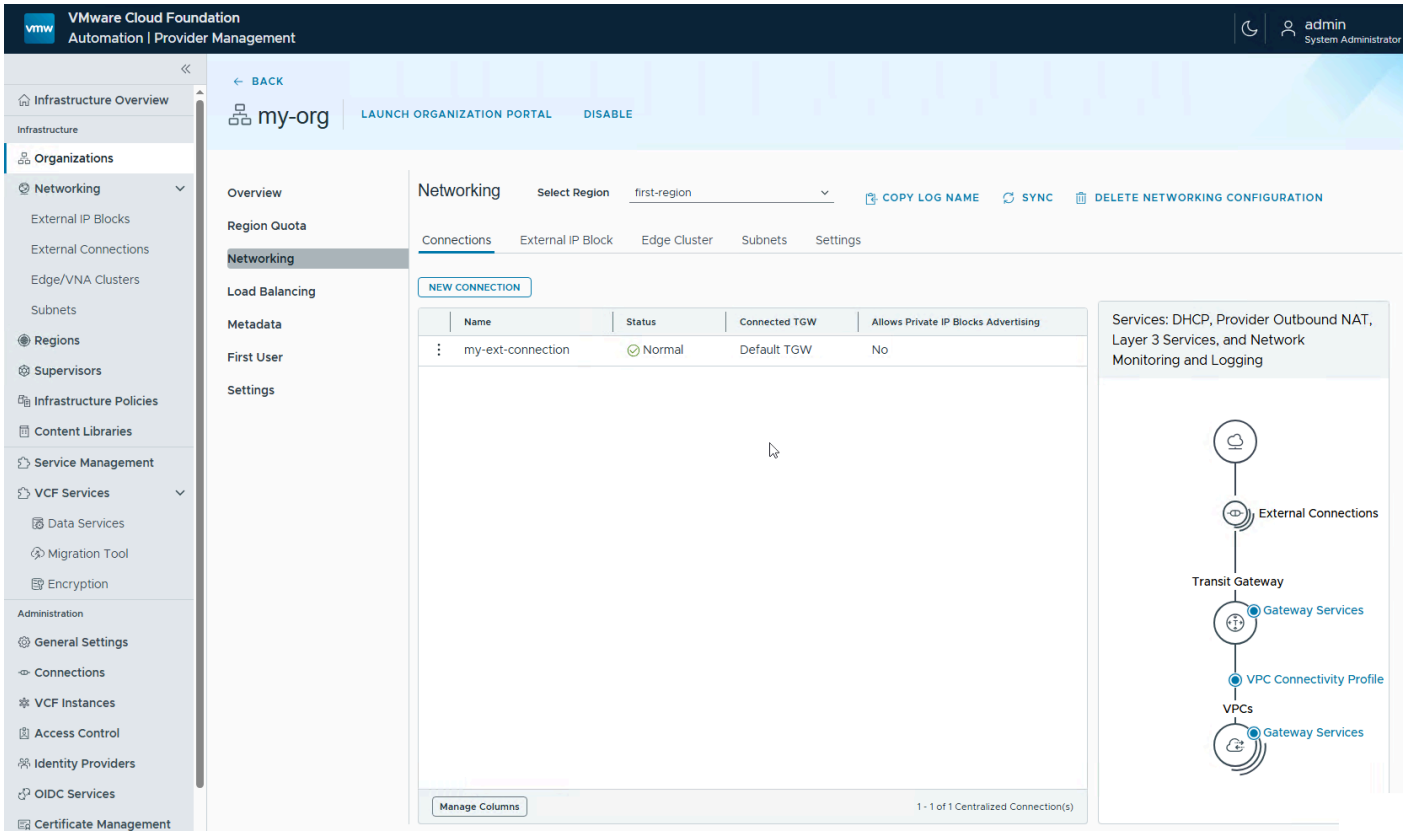
2. Deploy Supervisor instances with NSX VPC in the vSphere Client. You can also activate the Supervisor during workload domain deployment.

Figure 11: Deploying a Supervisor on One or More vSphere Clusters in the vSphere Client



3. Add an identity provider for your organizations.
4. Create organizations for the workload domains.

Figure 12: Managing Organizations in VCF Automation Provider Management



a. Create organizations for all applications for the workload domains with Supervisor instances.
b. Create an organization for VM applications for the workload domains that do not run Supervisor instances.

You can also share a workload domain between an All Apps Organization and a VM Apps Organization.

Design Blueprints

You can apply a pre-defined architecture for your specific business and technology requirements. See Infrastructure Modernizationdesign blueprints.

List of VCF Capabilities

For a detailed list of capabilities and links to the documentation for them, see VCF Capabilities by Job Function.

What to Do Next

Share the details of the organizations you created in VCF Automation provider management with the administrators for those organizations. Organization administrators can start distributing the infrastructure and configuring access management to the groups in their organizations by using the organization management functionality in VCF Automation. See Managing Organizations in VMware Cloud Foundation Automation and Managing Organizations for VM Apps in VMware Cloud Foundation Automation.

Getting Started with vSphere Foundation

You can start building a vSphere Foundation private cloud by deploying a new platform or converging your existing virtual infrastructure to vSphere Foundation. After you deploy VCF Operations, VCF management services, and license server using a new or existing vCenter instance, you can continue with adding vCenter instances to the cloud.

In this article:

- Deploy VCF Operations
- Add vCenter Instances to VCF Operations
- Complete the Configuration of vSphere Foundation
- What to Do Next

Deploy VCF Operations

The first step in building a vSphere Foundation platform is deploying VCF Operations, VCF management services, and the license server component in one of the following ways:

- Deploy a new vSphere Foundation platform.
- Deploy VCF Operations, VCF management services, and license server on an existing vCenter instance.

If you have an existing VMware Aria Operations and vCenter instance with managed vSphere clusters, upgrade VMware Aria Operations to VCF Operations 9.1 and vCenter to version 9.1.

Add vCenter Instances to VCF Operations

After you deploy or upgrade to VCF Operations, you can start adding vCenter instances with infrastructure running consumer workloads. See vSphere as a data source in VCF Operations.

Complete the Configuration of vSphere Foundation

After you deploy VCF Operations or add a vCenter instance to it, complete the platform's configuration by performing the following tasks. Assigning a license must be done right after you complete deployment.

- Assign a license.
- Configure software depot settings in VCF Operations
- Configure backup of the management components.
- Configure certificate management in VCF Operations.
- Deploy and configure vSphere Supervisor.

What to Do Next

If you want to run Kubernetes workloads, VMs, or vSphere Pods on vSphere, see Building Cloud Applications.